

Electrodynamics of spinless particles

Assume spinless electrons: $(\partial_\mu \partial^\mu + m^2)\phi = 0$

From classical electrodynamics: $p^\mu \rightarrow p^\mu + e A^\mu$

$$\Rightarrow i\partial^\mu \rightarrow i\partial^\mu + e A^\mu$$

$$A^\mu = (A^0, \vec{A})$$

EM potential

The K-G eqn becomes: $(\partial_\mu \partial^\mu + m^2)\phi + [ie(\partial_\mu A^\mu + A_\mu \partial^\mu) + e^2 A^2]\phi = 0$

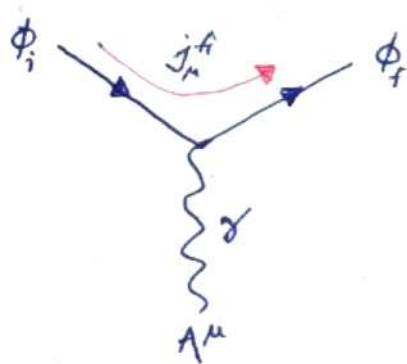
Since the electric charge is small ($\alpha \ll 1$), we can treat the potential term as a perturbation and ignore the order e^2 term.

$$\begin{aligned} T_{fi} &= -i \int \phi_f^*(x) V(x) \phi_i(x) d^4x \\ &= i \int \phi_f^* (ie(\partial_\mu A^\mu + A_\mu \partial^\mu)) \phi_i d^4x \\ &= -e \int [\phi_f^* (\partial_\mu \phi_i) - (\partial_\mu \phi_f^*) \phi_i] \cdot A^\mu d^4x \end{aligned}$$

$$\text{or } T_{fi} = -i \int j_\mu^{fi} A^\mu d^4x$$

where $j_\mu^{fi} = -ie(\phi_f^* (\partial_\mu \phi_i) - (\partial_\mu \phi_f^*) \phi_i) = \text{EM current.}$

Flipped the derivative and dropped the surface term.



A free electron interacts with the potential (a photon) and becomes free again

$$\phi_{i/f}(x) = N_{i/f} e^{-i p_{i/f} \cdot x}$$

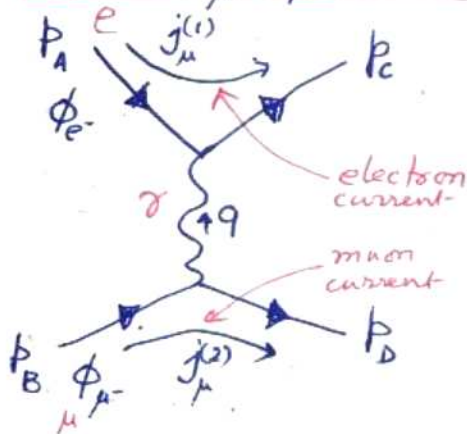
{ 4-vectors dot product }

$$j_\mu^{fi} = -e N_i N_f (p_{i\mu} + p_{f\mu}) e^{i(p_f - p_i) \cdot x}$$

EM current carried by the spinless electron

(Photon is the quantum of EM potential)

We want to calculate the probability of the scattering of spinless electrons off spinless muons (say)



Same calculation, only we need to identify the source of the EM potential with the muon current.

What is A^μ generated by j^μ ?

We need to rewrite Maxwell's eqns in terms of the potential

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

$$\vec{E} = -\vec{\nabla} A^0 - \frac{\partial \vec{A}}{\partial t}$$

$$A^\mu = (A^0, \vec{A})$$

Maxwell's eqns: ($\epsilon_0 = \mu_0 = 1$)

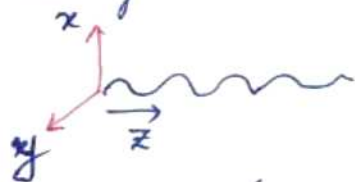
$$\boxed{\vec{\nabla} \cdot \vec{E} = \rho} \Rightarrow \vec{\nabla} \cdot \left(-\vec{\nabla} A^0 - \frac{\partial \vec{A}}{\partial t} \right) = j^0 \Rightarrow \boxed{\nabla^2 A^0 + \frac{\partial}{\partial t} (\vec{\nabla} \cdot \vec{A}) = -j^0}$$

$$\boxed{\vec{\nabla} \times \vec{B} - \frac{\partial \vec{E}}{\partial t} = \vec{j}} \Rightarrow \boxed{\left(-\nabla^2 + \frac{\partial^2}{\partial t^2} \right) \vec{A} + \vec{\nabla} \left(\vec{\nabla} \cdot \vec{A} + \frac{\partial A^0}{\partial t} \right) = \vec{j}}$$

Prove this

There is some redundancy in A^μ .

EM waves are transverse waves. Its polarisation ($\vec{E}$ or $\vec{B}$ field) is always $\perp$ to the direction of propagation.



A^μ has four degrees of freedom (4-components)
 $\Rightarrow$ two are unphysical! We can put arbitrary constraints to get rid of those. (Gauge Fixing)

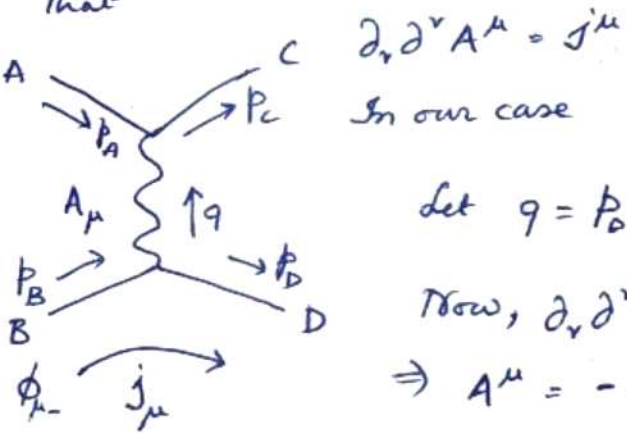
We choose 'Lorentz Gauge': $\boxed{\vec{\nabla} \cdot \vec{A} = -\frac{\partial A^0}{\partial t}}$

Maxwell's eqns become: $\boxed{\left(-\nabla^2 + \frac{\partial^2}{\partial t^2} \right) A^0 = j^0} \quad \boxed{\left(-\nabla^2 + \frac{\partial^2}{\partial t^2} \right) \vec{A} = \vec{j}}$

or, more compactly:

$$\boxed{\square^2 A^\mu \equiv \partial_\nu \partial^\nu A^\mu \equiv \left(\frac{\partial^2}{\partial t^2} - \nabla^2 \right) A^\mu = j^\mu}$$

So, if there is a current j^μ , it will produce a potential A^μ such that



$$\partial_\nu \partial^\nu A^\mu = j^\mu$$

In our case $j_{(2)}^\mu = -e N_D N_B (p_D^\mu + p_B^\mu) e^{i(p_D - p_B) \cdot x}$

Let $q = p_D - p_B$. $\therefore j_{(2)}^\mu = -e N_B N_D (p_B^\mu + p_D^\mu) e^{iq \cdot x}$

Now, $\partial_\nu \partial^\nu e^{iq \cdot x} = -q^2 e^{iq \cdot x}$

$\Rightarrow A^\mu = -\frac{j_{(2)}^\mu}{q^2}$ is a solution of the Maxwell's eqn.

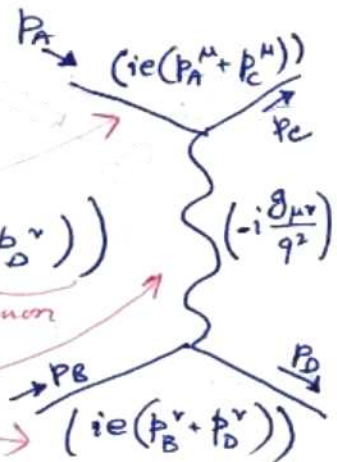
$\Rightarrow T_{fi} = -i \int j_\mu^{(1)} \left(-\frac{1}{q^2}\right) j_\mu^{(2)} d^4x$

slow this!

$= -i N_A N_B N_C N_D (2\pi)^4 \underbrace{\delta^{(4)}(p_D + p_C - p_A - p_B)}_{\text{Momentum conservation}} e i \mathcal{M}_0$

where

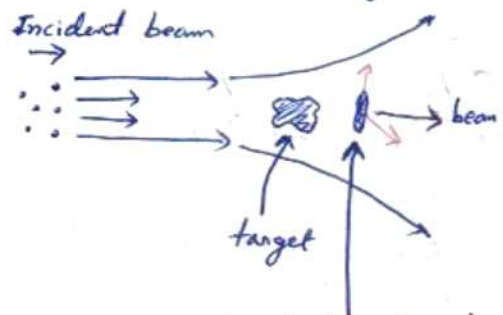
$-i \mathcal{M}_0 = \underbrace{(ie(p_A^\mu + p_C^\mu))}_{\text{spinless electron current}} \underbrace{\left(-i \frac{g_{\mu\nu}}{q^2}\right)}_{\text{photon propagator}} \underbrace{(ie(p_B^\nu + p_D^\nu))}_{\text{spinless muon current}}$



def. $\delta(x-x_0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ip(x-x_0)} dp$

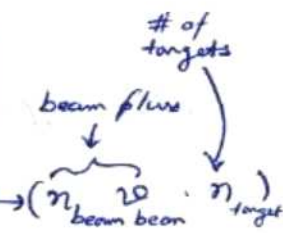
Now that we know T_{fi} , we can calculate the transition rate per unit volume $W_{fi} = \frac{|T_{fi}|^2}{TV}$

We want to connect this to scattering cross section which is a measure of the probability of the scattering process will occur.



⊥ projection of the obstacle $\sim$ prob. of scattering

$$\begin{aligned} \# \text{ of scattered particles } (n_s) \\ = \sigma \times (\text{initial flux}) \\ \text{(cross section)} \end{aligned}$$



Also,

$$n_s = W_{fi} \times (\text{number of final states})$$

We know the final state number from the 'particle in a box' problem in QM.

If there are N_v number of particles with momenta between $\vec{p}$ and $\vec{p} + d\vec{p}$ in a box of volume V ,

of possible final states / particle

$$= \frac{V d^3\vec{p}}{(2\pi)^3} \times \frac{1}{N_v} \Rightarrow \text{with this we can get the differential cross-section, do}$$

Let us take the volume V such that there are $2E$ number of particles
i.e., we fix normalization such that

$$\int_V p dV = N_V = 2E \quad \left[\rho = 2E/N \right] \leftarrow \phi = N e^{-i p \cdot x}$$

i.e., $N = \frac{1}{\sqrt{V}}$ check!

This implies

$$W_{fi} = \frac{|T_{fi}|^2}{TV} = \frac{(2\pi)^4 \delta^{(4)}(p_c + p_d - p_a - p_b) |\mathcal{M}_0|^2}{V^4}$$

or,

Now we ^{just} need to estimate the initial flux in the Lab frame (where target is fixed)

of beam particles passing through unit area per unit time is $|\vec{v}_A| \frac{2E_A}{V}$

there are going to find $\frac{2E_B}{V}$ target particles per unit volume.

$$\therefore \text{initial flux} = |\vec{v}_A| \frac{2E_A}{V} \frac{2E_B}{V}$$

rel. velocity of A (beam particles)
w.r.t. B (target particles)

Putting all together

$$\therefore d\sigma = \left[\frac{1}{|\vec{v}_A| \frac{4E_A E_B}{V^2}} \right] \left[\frac{(2\pi)^4 \delta^{(4)}(p_c + p_d - p_a - p_b) |\mathcal{M}_0|^2}{V^4} \right] \left[\frac{V d^3 \vec{p}_c}{2E_c (2\pi)^3} \cdot \frac{V d^3 \vec{p}_d}{2E_d (2\pi)^3} \right]$$

(All V dependence cancels out. We can take $V=1$)

The differential cross section for final-state particles have momentum between $(\vec{p}_c \text{ and } \vec{p}_c + d\vec{p}_c)$ and $(\vec{p}_D \text{ and } \vec{p}_D + d\vec{p}_D)$ is

$$d\sigma = \frac{|\mathcal{M}|^2}{F} dQ$$

square of the Lorentz-invariant amplitude
 Lorent invariant phase-space factor

Flux

$$\text{where } dQ = (2\pi)^4 \delta^{(4)}(\underbrace{\vec{p}_c + \vec{p}_D}_{\text{Final mom.}} - \underbrace{(\vec{p}_A + \vec{p}_B)}_{\text{initial momentum}}) \underbrace{\frac{d^3\vec{p}_c}{(2\pi)^3 2E_c}}_{\text{For every final-state particle}} \underbrace{\frac{d^3\vec{p}_D}{(2\pi)^3 2E_D}}_{\text{For every final-state particle}}$$

In general

$$dQ = (2\pi)^4 \delta^{(4)}\left(\sum_f \vec{p}_f - \sum_i \vec{p}_i\right) \prod_f \frac{d^3\vec{p}_f}{(2\pi)^3 2E_f}$$

$$\Rightarrow F = |\vec{V}_A - \vec{V}_B| 2E_A \cdot 2E_B$$

For back-to-back scattering:

$$F = 4 (|\vec{p}_A| E_B + |\vec{p}_B| E_A) \quad \begin{matrix} \text{A} \longrightarrow & \longleftarrow \text{B} \end{matrix}$$

$$= 4 \sqrt{(\vec{p}_A \cdot \vec{p}_B)^2 - m_A^2 m_B^2}$$

Show

(Invariant under boost along the beam axis)

$\frac{d^3\vec{p}}{(2\pi)^3 2E}$ is Lorentz-invariant.

Notice, d^4p is Lorentz-invariant $[d^4p \rightarrow |\det \Lambda| d^4p = d^4p]$
outgoing particles are real, i.e., not virtual.

Now, $\frac{d^4p}{(2\pi)^3} \delta(p_0^2 - \vec{p}^2 - m^2) \Theta(p_0)$ is also Lorentz invariant.

$\Theta(x) = 1$ if $x > 0$
 $= 0$ otherwise

Heaviside-theta fn

$$= \frac{d^3\vec{p}}{(2\pi)^3} dp_0 \delta(p_0^2 - \vec{p}^2 - m^2) \Theta(p_0)$$

$$= \frac{d^3\vec{p}}{(2\pi)^3} dp_0 \frac{1}{2p_0} \left[\delta(p_0 - \sqrt{\vec{p}^2 + m^2}) + \delta(p_0 + \sqrt{\vec{p}^2 + m^2}) \right] \Theta(p_0)$$

$$= \frac{d^3\vec{p}}{(2\pi)^3 2E} \text{ where } E = \sqrt{\vec{p}^2 + m^2}$$

Formula:

$$\int_{-\infty}^{\infty} \delta(g(x)) f(x) dx = \sum_i \frac{f(x_i)}{\left| \frac{d}{dx} g(x) \right|_{x=x_i}} \text{ where } g(x_i) = 0$$

[The theta fn ensures ^{that} we take the +ve energy solution]

We can express the differential cross section in the centre-of-mass (com) frame in a nice and compact way.

In the COM frame: $dQ = \frac{1}{(4\pi)^2} \frac{d^3\vec{p}_c}{2E_c} \frac{d^3\vec{p}_D}{2E_D} \delta^{(4)}(p_c + p_D - p_A - p_B)$

$$= \frac{1}{(4\pi)^2} \frac{d^3\vec{p}_c}{2E_c} \frac{1}{2E_D} \delta(E_c + E_D - E_A - E_B)$$

[we have performed the integrals over $\vec{p}_D$ and set $\vec{p}_D = -\vec{p}_c$ in the form]

Let $s = (E_A + E_B)^2$ and $\vec{p}_c = \vec{p}_f$

$$\therefore dQ = \frac{1}{4\pi^2} \frac{|\vec{p}_f|^2 d|\vec{p}_f| d\Omega_f}{4E_c E_D} \delta(\sqrt{s} - E_c - E_D)$$

The delta fn. will set $\sqrt{s} = E_c + E_D = \sqrt{m_c^2 + |\vec{p}_f|^2} + \sqrt{m_D^2 + |\vec{p}_f|^2}$

$$\therefore \frac{d\sqrt{s}}{d|\vec{p}_f|} = |\vec{p}_f| \left(\frac{1}{E_c} + \frac{1}{E_D} \right) \Rightarrow d|\vec{p}_f| = \frac{E_c E_D d\sqrt{s}}{|\vec{p}_f| (E_c + E_D)}$$

Hence, $dQ = \frac{1}{4\pi^2} \frac{|\vec{p}_f| d\sqrt{s}}{4(E_D + E_c)} d\Omega_f \delta(\sqrt{s} - E_c - E_D)$

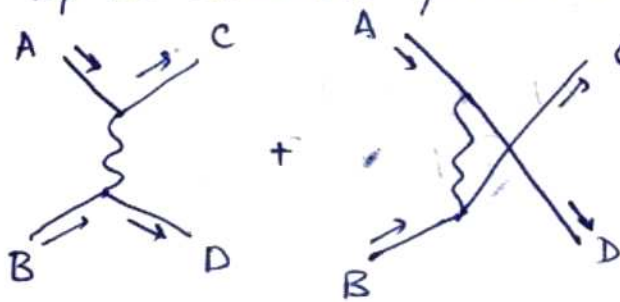
$$= \frac{1}{4\pi^2} \frac{|\vec{p}_f|}{4\sqrt{s}} d\Omega_f \quad [\text{we have performed the integral over } \sqrt{s}]$$

In the COM/CM frame

$$F = 4(|\vec{F}_i|E_A + |\vec{F}_i|E_B) = 4|\vec{F}_i|\sqrt{s} \quad \text{where } |\vec{F}_i| = |\vec{F}_A| = |\vec{F}_B|$$

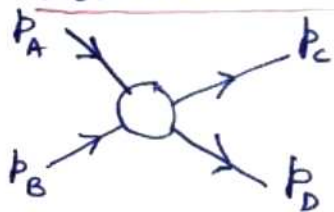
$$\therefore \left. \frac{d\sigma}{d\Omega} \right|_{\text{CM}} = \frac{1}{64\pi^2 s} \frac{|\vec{F}_f|}{|\vec{F}_i|} |\mathcal{M}|^2$$

If we consider spinless e^-e^- scattering



$$-ie\gamma_{\mu}e^{-} = -i \left[- \frac{e^2 (p_A^{\mu} + p_C^{\mu})(p_{B\mu} + p_{D\mu})}{(p_D - p_B)^2} - \frac{e^2 (p_A^{\mu} + p_D^{\mu})(p_{B\mu} + p_{C\mu})}{(p_C - p_B)^2} \right]$$

Mandelstam Variables:



$$s = (p_A + p_B)^2 = (p_C + p_D)^2$$

$$t = (p_A - p_C)^2 = (p_B - p_D)^2$$

$$u = (p_A - p_D)^2 = (p_B - p_C)^2$$

[Note in the CM frame $s = (E_A + E_B)^2 = (E_C + E_D)^2$]

These scalars satisfy $s+t+u = \sum_{i=A,B,C,D} m_i^2 + 2p_A \cdot p_B - 2p_A \cdot p_C - 2p_A \cdot p_D + 2p_A^2$ (11)

$$= \sum_i m_i^2 + \underbrace{2p_A \cdot (p_A + p_B - p_C - p_D)}_{=0}$$

or $\boxed{s+t+u = \sum_{i=A,B,C,D} m_i^2}$

In terms of these variables

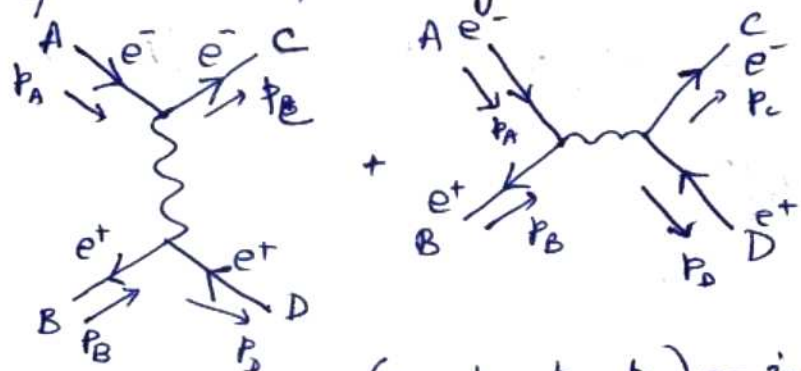
$$\mathcal{M}_{e^-e^-} = -e^2 \frac{(p_A + p_C) \cdot (p_B + p_D)}{t} - e^2 \frac{(p_A + p_D) \cdot (p_B + p_C)}{u}$$

$$\begin{aligned} \text{Now, } (p_A + p_C) \cdot (p_B + p_D) &= (2p_A + p_B - p_D) \cdot (p_B + p_D) \\ &= 2p_A \cdot p_B + p_B^2 - p_D^2 + 2p_A \cdot p_D \\ &= (p_A + p_B)^2 - (p_A - p_D)^2 \\ &= s - u \end{aligned}$$

Similarly, $(p_A + p_D) \cdot (p_B + p_C) = s - t$

$$\therefore \boxed{\mathcal{M}_{e^-e^-} = +e^2 \left[\frac{s-s}{u} + \frac{u-s}{t} \right]}$$

spinless e^+e^- scattering:



$$[p_D \leftrightarrow -p_B]$$

$$-i\mathcal{M}_{e^+e^-}(p_A, p_B, p_C, p_D) = -ie\mathcal{M}_{e^+e^-}(p_A, -p_D, p_C, -p_B)$$

$$= -i \left(e^2 \frac{(p_A + p_C) \cdot (-p_D - p_B)}{(p_D - p_B)^2} - e^2 \frac{(p_A - p_B) \cdot (-p_D - p_C)}{(p_C + p_D)^2} \right)$$

$$= -ie^2 \left(\frac{s-u}{t} + \frac{t-u}{s} \right)$$

$$\mathcal{M}_{e^+e^-}(s, t, u) = \mathcal{M}_{e^+e^-}(u, t, s)$$

We can now connect the differential scattering cross section for the $e^- \mu^- \rightarrow e^- \mu^-$ process to measurable quantities. (13)

Let's say both the electron and muon have high energies
 $\Rightarrow m_e \simeq m_\mu \simeq 0$

In the CM frame

$$\begin{aligned} p_A &= E, \vec{p} & , & & p_C &= E', \vec{p}' \\ p_B &= E, -\vec{p} & , & & p_D &= E', -\vec{p}' \end{aligned}$$

$$\text{let } \hat{p} \cdot \hat{p}' = \cos \theta$$

We can show

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{CM}} = \frac{\alpha^2}{4s} \left(\frac{3 + \cos \theta}{1 - \cos \theta} \right)$$

(high energy)

Home Work.